

Lecture 3

Symplectic manifolds

Objectives. *Lecture 1* gave a bracket and *Lecture 2* gave forms. They are the same structure, and this lecture names it: a closed, non-degenerate 2-form ω . Naming it buys three things the rest of Part I spends. Darboux's theorem says that all the interesting content of a symplectic manifold is global, which is exactly what the Liouville–Arnold theorem of *Lecture 4* exploits; Lagrangian submanifolds are the shape its invariant tori will turn out to have; and the canonical 1-form $p \cdot dq$ is what the action variables of *Lecture 5* are built from.

3.1 The symplectic form

Lecture 4's first step is that a regular level set of n independent commuting integrals is *Lagrangian* (§ 3.6), and its difficulty is that its conclusion is global, because locally there is nothing to prove (§ 3.3). *Lecture 5* needs $p \cdot dq$ to be a genuine 1-form on phase space (§ 3.7) and generating functions to produce canonical transformations (§ 3.4). One piece of notation first: a 2-form eats two vectors, and feeding it one leaves a 1-form.

Definition 3.1 (Interior product). For a 2-form β and a vector v at the same point, $\iota_v \beta$ is the 1-form $w \mapsto \beta(v, w)$. The slot that v fills is the *first*.

Definition 3.2 (Symplectic form, symplectic manifold). A 2-form ω on a manifold $\mathcal{M}$ is *symplectic* if

1. it is *closed*: $d\omega = 0$; and
2. it is *non-degenerate*: at each point, if $\omega(v, w) = 0$ for every w , then $v = 0$.

The pair $(\mathcal{M}, \omega)$ is a *symplectic manifold*.

Condition 2 says that the linear map $v \mapsto \iota_v \omega$ from $T_p \mathcal{M}$ to $T_p^* \mathcal{M}$ has zero kernel. The two spaces have the same dimension, so

$$v \longmapsto \iota_v \omega \quad \text{is an isomorphism } T_p \mathcal{M} \xrightarrow{\sim} T_p^* \mathcal{M}. \quad (3.1)$$

A *symplectic form* converts vectors into covectors and back, with no choices made — the job an inner product does in Euclidean space, done by an antisymmetric object.

Example 3.3 (The standard symplectic form). On $\mathbb{R}^{2n}$ with coordinates $q_1, \dots, q_n, p_1, \dots, p_n$,

$$\omega = \sum_{i=1}^n dq_i \wedge dp_i, \quad \omega(v, w) = \sum_{i=1}^n (v^{q_i} w^{p_i} - v^{p_i} w^{q_i}), \quad (3.2)$$

by (2.5). Geometrically, $\omega(v, w)$ is the sum over i of the signed areas of the shadows of the parallelogram on v, w in the (q_i, p_i) -planes; for $n = 1$ it is the signed area itself (Figure 3.1). It pairs each position only with its own momentum.

In matrix form, with vectors written as columns in the order (q, p) ,

$$\omega(v, w) = v^T \Omega w, \quad \Omega = \begin{pmatrix} 0 & \mathbb{1}_n \\ -\mathbb{1}_n & 0 \end{pmatrix}. \quad (3.3)$$

It is closed because its coefficients are constant, and non-degenerate because $\det \Omega = 1$.

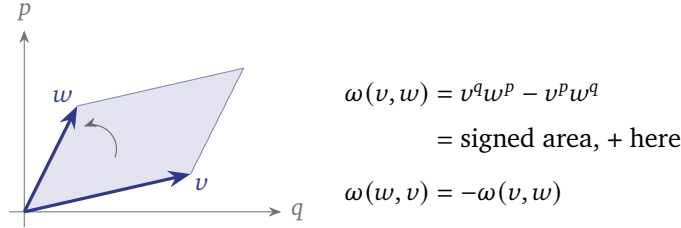


Figure 3.1: On one degree of freedom ω is the signed area in the (q, p) -plane — Figure 2.5 with q, p for x, y . For n degrees of freedom it adds n such areas; Figure 3.5 shows $n = 2$.

Proposition 3.4 (Phase space has even dimension). *A non-degenerate 2-form exists only on a manifold of even dimension.*

Proof. At a point, write ω as an $m \times m$ matrix W in some basis. It is antisymmetric, so $\det W = \det W^T = \det(-W) = (-1)^m \det W$. For m odd this forces $\det W = 0$, so W has a kernel and ω is degenerate. ■

The sign test, on one degree of freedom

The wedge order $dq \wedge dp$ is not a matter of taste. Ask which X satisfies $\iota_X \omega = dH$ on $\mathbb{R}^2$. By Definition 3.1 and (2.5),

$$\iota_X(dq \wedge dp) = dq(X) dp - dp(X) dq = X^q dp - X^p dq,$$

and matching with $dH = H_q dq + H_p dp$ gives

$$X^q = \frac{\partial H}{\partial p}, \quad X^p = -\frac{\partial H}{\partial q} : \quad (3.4)$$

Hamilton's equations, both signs right; for the pendulum $H = \frac{1}{2}p^2 - \cos q$ this is $\dot{q} = p$, $\dot{p} = -\sin q$. Had we written $dp \wedge dq$, the same computation would return $X^q = -H_p$ and $X^p = +H_q$ — both signs wrong. That is the whole reason for the order.

What each condition is for

Non-degeneracy makes $\iota_X \omega = dH$ uniquely solvable for every H , by (3.1). Without it a function need not determine a flow: the closed form $q dq \wedge dp$ on $\mathbb{R}^2$ degenerates on the line $q = 0$, where $\iota_X \omega = 0$ for every X and $\iota_X \omega = dH$ has no solution wherever $dH \neq 0$. *Closedness* is what makes the bracket built from ω satisfy the Jacobi identity (Theorem 3.8).

3.2 $\iota_{X_f} \omega = df$: one structure, two languages

Lecture 1 defined X_f from the bracket, by $X_f h = \{h, f\}$ (Definition 1.14). On a symplectic manifold there is a second route, which uses no bracket at all.

Definition 3.5 (Hamiltonian vector field, from ω). On a symplectic manifold (M, ω) , the Hamiltonian vector field of a smooth function f is the unique vector field X_f with

$$\iota_{X_f} \omega = df. \quad (3.5)$$

It exists and is unique at each point by (3.1), and it is smooth because ω and f are.

Proposition 3.6 (The two definitions agree). In the coordinates of (3.2), $X_f = \sum_i (\frac{\partial f}{\partial p_i} \partial_{q_i} - \frac{\partial f}{\partial q_i} \partial_{p_i})$, which is [Lecture 1](#)'s X_f exactly. Consequently

$$\omega(X_f, X_g) = \{f, g\}. \quad (3.6)$$

Proof. The first statement is the sign test (3.4), run once for each pair (q_i, p_i) with f in place of H ; the pairs do not interact, because ω does not. For the second,

$$\omega(X_f, X_g) = (\iota_{X_f} \omega)(X_g) = df(X_g) = X_g f = \{f, g\},$$

using in turn [Definition 3.1](#), (3.5), (2.2), and [Lecture 1](#)'s definition $X_g f = \{f, g\}$ (1.13). ■

On [Lecture 1](#)'s test case, $f = q^2/2$ and $g = p^2/2$ with $X_f = -q \partial_p$ and $X_g = p \partial_q$ (1.15),

$$\omega(X_f, X_g) = dq(X_f) dp(X_g) - dp(X_f) dq(X_g) = 0 \cdot 0 - p \cdot (-q) = qp = \{f, g\}.$$

Example 3.7 (Angular momentum generates rotations). Solve (3.5) a second time, for $\ell = q_1 p_2 - q_2 p_1$ on $\mathbb{R}^4$. The two sides are

$$\iota_X \omega = \sum_i (X^{q_i} dp_i - X^{p_i} dq_i), \quad d\ell = p_2 dq_1 - p_1 dq_2 - q_2 dp_1 + q_1 dp_2,$$

and matching coefficients gives

$$X_\ell = -q_2 \partial_{q_1} + q_1 \partial_{q_2} - p_2 \partial_{p_1} + p_1 \partial_{p_2} :$$

a rotation of (q_1, q_2) and, at the same rate, of (p_1, p_2) . This is the flow that [Lecture 1](#)'s one-line Noether theorem was about: ℓ is conserved exactly when H is invariant under it.

Any symplectic form gives a bracket

Turn (3.6) round and make it a definition: on any symplectic manifold, put

$$\{f, g\} := \omega(X_f, X_g) = df(X_g). \quad (3.7)$$

It is bilinear and antisymmetric, and satisfies Leibniz because $d(fg) = f dg + g df$. What is left is Jacobi, and this is where closedness enters.

Theorem 3.8 (Jacobi is $d\omega = 0$). Let ω be a non-degenerate 2-form, with X_f and $\{f, g\}$ defined by (3.5) and (3.7). Then for all f, g, h ,

$$\{f, \{g, h\}\} + \{g, \{h, f\}\} + \{h, \{f, g\}\} = d\omega(X_f, X_g, X_h). \quad (3.8)$$

Hence the bracket satisfies the Jacobi identity if and only if $d\omega = 0$.

The proof, in the reading, is [Lecture 1](#)'s computation with non-constant coefficients: the terms in which a derivative lands on a coefficient no longer vanish, and they assemble into exactly $d\omega$.

Proof (★). Work in any chart $z^1, \dots, z^{2n}$, and write $\omega_{ab} = \omega(\partial_a, \partial_b)$ and $W = (\omega_{ab})$, an antisymmetric matrix, invertible by non-degeneracy. Set $J = -W^{-1}$, also antisymmetric.

Step 1: the bracket in coordinates. Condition (3.5) reads $\sum_a X_g^a \omega_{ab} = \partial_b g$ for each b , whose solution is $X_g^a = \sum_b J^{ab} \partial_b g$. So

$$\{f, g\} = df(X_g) = \sum_{a,b} \partial_a f J^{ab} \partial_b g, \quad J^{ab} = \{z^a, z^b\}.$$

This is [Lecture 1](#)'s formula (1.11) — except that J is now allowed to vary from point to point.

Step 2: what survives in the Jacobi sum. Expand the cyclic sum. The terms carrying a second derivative of one of f, g, h cancel exactly as in the proof of [Proposition 1.12](#): that argument used only that J is antisymmetric at each point, never that it is constant. What is new are the terms in which a derivative lands on J . Collecting them, with repeated indices summed,

$$\{f, \{g, h\}\} + \text{cyclic} = \partial_a f \partial_c g \partial_d h S^{acd}, \quad S^{acd} = J^{ab} \partial_b J^{cd} + J^{cb} \partial_b J^{da} + J^{db} \partial_b J^{ac}.$$

Step 3: trade ∂J for $\partial \omega$. Differentiate $JW = -\mathbb{1}$: $(\partial_b J)W + J \partial_b W = 0$, so $\partial_b J = -J(\partial_b W)W^{-1} = J(\partial_b W)J$. Substituting into each of the three terms of S and relabelling dummy indices so that every term carries the same three factors of J ,

$$S^{acd} = -J^{ab} J^{ce} J^{df} (\partial_b \omega_{ef} + \partial_e \omega_{fb} + \partial_f \omega_{be}).$$

Step 4: recognise $d\omega$. By [Definition 2.11](#), the coefficient of $dz^b \wedge dz^e \wedge dz^f$ in $d\omega$ collects exactly the three terms in the bracket, so it is the component $(d\omega)_{bef}$. And since $X_f^b = J^{ba} \partial_a f = -J^{ab} \partial_a f$, each of the three factors $\partial f J$ in $\partial_a f \partial_c g \partial_d h S^{acd}$ is minus a component of a Hamiltonian vector field. Three minus signs and the one in front of S make a plus:

$$\partial_a f \partial_c g \partial_d h S^{acd} = (d\omega)_{bef} X_f^b X_g^e X_h^f = d\omega(X_f, X_g, X_h),$$

the last step being the determinant formula (2.8) for a 3-form evaluated on three vectors. That is (3.8).

If $d\omega = 0$ the right-hand side vanishes, and Jacobi holds. Conversely, if Jacobi holds for all f, g, h , take them among the coordinate functions: the fields $X_{z^a} = J e_a$ are the columns of the invertible matrix J , so they span every tangent space, and a 3-form that vanishes on every triple from a basis is zero. ■

Reading

Example 3.9 (A non-degenerate form that is not closed). On $\mathbb{R}^4$ take $\omega' = (1 + q_2^2) dq_1 \wedge dp_1 + dq_2 \wedge dp_2$. It is non-degenerate everywhere, and not closed: $d\omega' = 2q_2 dq_2 \wedge dq_1 \wedge dp_1$. Its bracket looks perfectly reasonable — $\{q_1, p_1\} = 1/(1 + q_2^2)$, $\{q_2, p_2\} = 1$, the rest zero — and it is not a Poisson bracket:

$$\{q_1, \{p_1, p_2\}\} + \{p_1, \{p_2, q_1\}\} + \{p_2, \{q_1, p_1\}\} = \frac{2q_2}{(1 + q_2^2)^2} \neq 0.$$

[Lecture 1](#) proved the engine $X_{\{f,g\}} = -[X_f, X_g]$ from the Jacobi identity alone ([Theorem 1.15](#), second proof). So the engine runs on any closed non-degenerate 2-form, and on nothing less: that is the sense in which [Lecture 1](#)'s bracket and [Lecture 2](#)'s forms are one structure.

3.3 Darboux's theorem

Theorem 3.10 (Darboux · sketched). Every point of a $2n$ -dimensional symplectic manifold (M, ω) has a chart $q_1, \dots, q_n, p_1, \dots, p_n$ around it in which $\omega = \sum_i dq_i \wedge dp_i$. (da Silva, §8.1; Arnold, §43.)

Such a chart is a *Darboux chart*, or *canonical coordinates*. So any statement about ω alone — or about the bracket and the fields X_f built from it — that is proved in q 's and p 's holds near every point of every symplectic manifold.

What “no local invariants” means — and what it does not

Compare a Riemannian metric, which measures lengths and angles. A sphere of radius R has curvature $1/R^2$ at every point, and curvature is a *local invariant*: no chart, however small, makes the sphere's metric into the flat $dx^2 + dy^2$. Darboux's theorem says a symplectic form has nothing of the kind (Figure 3.2): every small patch of every $2n$ -dimensional symplectic manifold is indistinguishable from every other, and the dimension is the only local invariant there is.

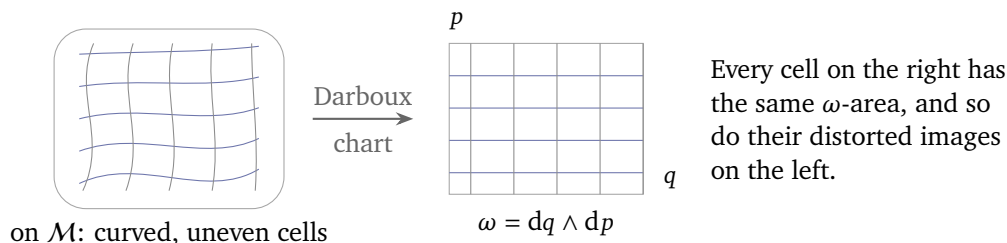


Figure 3.2: A Darboux chart straightens a symplectic form. Contrast a curved surface, where no chart can make the metric flat even on a tiny patch.

It is easy to hear “no local invariants” as “nothing interesting happens”, and that is the opposite of the point: everything interesting is *global*. It lives in loops that do not bound, in regions of finite size, in how charts are glued together — never at a point. The actions of [Lecture 5](#) are integrals over loops, and the tori of [Lecture 4](#) are compact; neither is visible in a Darboux chart.

Why it is true: Moser's idea

At a single point, linear algebra already makes ω standard (the reading proves this). Call that constant form ω_0 and join it to ω by the straight line $\omega_t = \omega_0 + t(\omega - \omega_0)$, $0 \leq t \leq 1$. Near the point each ω_t is non-degenerate, and $\omega - \omega_0$ is closed, hence $\omega - \omega_0 = d\sigma$ by the Poincaré lemma ([Theorem 2.15](#)). Now look for a flow ψ_t that pulls every ω_t back to ω_0 . Its velocity V_t is forced: $\iota_{V_t}\omega_t = -\sigma$, which non-degeneracy solves. That this V_t does the job is one differentiation, which uses $d\omega_t = 0$ and a formula for differentiating a form along a flow that we cite (reading). Then ψ_1 carries ω to ω_0 , and its coordinates are a Darboux chart.

At a point it is linear algebra; near a point it is Moser

Lemma 3.11 (Linear Darboux). *Let ω be a non-degenerate antisymmetric bilinear form on a vector space V . Then V has a basis $e_1, \dots, e_n, f_1, \dots, f_n$ with $\omega(e_i, f_j) = \delta_{ij}$ and $\omega(e_i, e_j) = \omega(f_i, f_j) = 0$. In particular $\dim V = 2n$.*

Proof. Pick any $e_1 \neq 0$. By non-degeneracy some f_1 has $\omega(e_1, f_1) \neq 0$; rescale it to 1. On $U = \text{span}(e_1, f_1)$ the form is non-degenerate, and its ω -complement $U^\omega = \{v : \omega(v, e_1) = \omega(v, f_1) = 0\}$ has dimension $\dim V - 2$ and meets U only in 0. So $V = U \oplus U^\omega$, and ω is still non-degenerate on U^ω : a vector there pairing to zero with all of U^ω also pairs to zero with U , hence with V . Repeat inside U^ω . ■

Lemma 3.11 makes ω standard at one point by a linear change of coordinates. A metric can be pushed a little further — suitable coordinates make it Euclidean at a point to first order — but no further: at second order its curvature is in the way. A symplectic form can be made standard on a whole neighbourhood, and that is Theorem 3.10. The proof spends closedness in exactly one place.

Proof of Theorem 3.10 (sketch: Moser's trick) (★). Work in a chart around a point z_0 in which, by Lemma 3.11, $\omega_1 := \omega$ agrees at z_0 with the constant form $\omega_0 = \sum dq_i \wedge dp_i$. Join them by $\omega_t = \omega_0 + t(\omega_1 - \omega_0)$, $t \in [0, 1]$. Each ω_t is closed, and equals ω_0 at z_0 , so it is non-degenerate on a small ball around z_0 . On that ball $\omega_1 - \omega_0$ is closed, hence exact by the Poincaré lemma (Theorem 2.15): $\omega_1 - \omega_0 = d\sigma$, and σ can be chosen to vanish at z_0 by subtracting its constant part. Define a time-dependent vector field V_t by $\iota_{V_t}\omega_t = -\sigma$; it exists by non-degeneracy and vanishes at z_0 , so its flow ψ_t fixes z_0 and is defined near z_0 up to $t = 1$. The one tool used without proof is Cartan's formula for the rate of change of a form along a flow (da Silva, §6.1), which gives

$$\frac{d}{dt} \psi_t^* \omega_t = \psi_t^* (d \iota_{V_t} \omega_t + \iota_{V_t} d\omega_t + \omega_1 - \omega_0) = \psi_t^* (-d\sigma + 0 + d\sigma) = 0.$$

Closedness is spent in the middle term. So $\psi_1^* \omega_1 = \omega_0$, and the coordinates q, p composed with ψ_1^{-1} are a Darboux chart for ω . ■

The use Lectures 4 and 5 make of it

Theorem 3.12 (Darboux, adapted to commuting functions · cited). *Let $F_1, \dots, F_n$ be functions with $\{F_i, F_j\} = 0$ and $dF_1, \dots, dF_n$ linearly independent at a point. Near that point there are functions $\psi_1, \dots, \psi_n$ such that (ψ, F) is a Darboux chart: $\omega = \sum_i d\psi_i \wedge dF_i$. (Bolsinov–Fomenko, Lemma 1.1, stated there in their convention $\omega = \sum dp \wedge dq$.)*

In such a chart the F_i are momenta and the ψ_i positions, so by Proposition 3.6 $X_{F_j} = \partial/\partial\psi_j$: the commuting flows are translations at unit speed. That is the whole of the Liouville–Arnold theorem locally, and it costs nothing — as Babelon, Bernard and Talon remark (ch. 2, Remark 2), near any point where $dH \neq 0$ Darboux gives canonical coordinates with $H = P_1$, so every Hamiltonian is “integrable” on a small enough patch.

So Lecture 4 is not trivial. Its question is whether the ψ_i continue over a whole level set, and what they do when they are continued: they come back to themselves, which makes them angles and the level set a torus. Action-angle coordinates are an adapted Darboux

chart on a whole neighbourhood of a torus, and the global part is what [Lectures 4 and 5](#) add.

3.4 Canonical transformations and generating functions

Definition 3.13 (Symplectomorphism). A diffeomorphism $F: (\mathcal{M}, \omega) \rightarrow (\mathcal{M}', \omega')$ is a *symplectomorphism* if $F^*\omega' = \omega$.

In canonical coordinates, with F written as $(Q, P) = F(q, p)$, the condition is

$$\sum_i dQ_i \wedge dP_i = \sum_i dq_i \wedge dp_i, \quad (3.9)$$

where on the left Q and P are regarded as functions of (q, p) and substituted — pullback is substitution (§ 2.5). It is a condition you already know under another name, since the matrix of the pulled-back form is $D^T \Omega D$, with D the Jacobian of F .

Proposition 3.14 (Symplectomorphisms are the canonical transformations). $(Q, P) = F(q, p)$ satisfies (3.9) if and only if the new coordinates have canonical brackets, $\{Q_i, P_j\} = \delta_{ij}$ and $\{Q_i, Q_j\} = \{P_i, P_j\} = 0$.

Proof (★). By (3.3) the pulled-back form has matrix $D^T \Omega D$, so (3.9) says $D^T \Omega D = \Omega$. On the other hand, by the chain rule and $\{z^a, z^b\} = \Omega^{ab}$, the brackets of the new coordinates form the matrix $D \Omega D^T$, and canonical brackets say $D \Omega D^T = \Omega$. The two conditions are equivalent: either one forces $\det D = \pm 1$, and inverting $D^T \Omega D = \Omega$ gives $D^{-1} \Omega^{-1} D^{-T} = \Omega^{-1}$, which with $\Omega^{-1} = -\Omega$ is $D \Omega D^T = \Omega$; the converse is the same computation backwards. ■

For one degree of freedom $dQ \wedge dP = (\det D) dq \wedge dp$, so a *canonical transformation of the plane is exactly an area- and orientation-preserving map*. To check the shear $(q, p) \mapsto (q + p^2, p)$, pull back: $d(q + p^2) \wedge dp = dq \wedge dp + 2p dp \wedge dp = dq \wedge dp$. The stretch $(q, p) \mapsto (2q, p)$ fails, with $\{2q, p\} = 2$. For $n \geq 2$ the condition is much stronger than preserving volume (§ 3.5).

Proposition 3.15 (Hamiltonian flows are symplectic · proved on $\mathbb{R}^{2n}$; modulo Darboux in general). For every t and every point at which the flow φ_t of X_H is defined, $\varphi_t^* \omega = \omega$.

Proof. First on $\mathbb{R}^{2n}$ with the standard form. By [Proposition 3.6](#), $X_H = \Omega \nabla H$ in the (q, p) ordering. The Jacobian $D(t)$ of φ_t at a point therefore obeys the linearised equation $\dot{D} = A(t)D$, $D(0) = \mathbb{1}$, with $A = \Omega S$ and S the (symmetric) Hessian of H along the trajectory. Then

$$\frac{d}{dt}(D^T \Omega D) = D^T (A^T \Omega + \Omega A) D = D^T (S \Omega^T \Omega + \Omega^2 S) D = D^T (S - S) D = 0,$$

because $\Omega^T \Omega = \mathbb{1}$ and $\Omega^2 = -\mathbb{1}$. So $D^T \Omega D = \Omega$ for all t , which is the claim on $\mathbb{R}^{2n}$; the reduction of a general symplectic manifold to this case is in the reading. ■

Reading

On a general symplectic manifold the trajectory may leave any one chart, but $\varphi_{t+s} = \varphi_s \circ \varphi_t$ gives

$$\frac{d}{dt} \varphi_t^* \omega = \varphi_t^* \left(\frac{d}{ds} \Big|_{s=0} \varphi_s^* \omega \right),$$

and the bracket is a rate of change at a single point, which may be computed in a Darboux chart there ([Theorem 3.10](#)): it is the computation above at $t = 0$, where $D = \mathbb{1}$, and it vanishes.

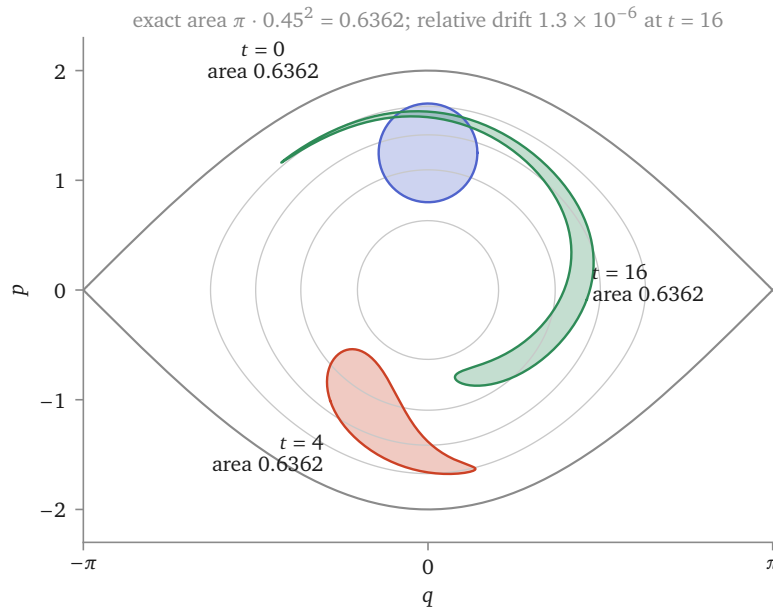


Figure 3.3: A symplectomorphism distorting a region and preserving its area: a disc of librating pendulum states carried by the flow for times 0, 4, 16. Outer orbits are slower ([Lecture 5](#)), so the disc shears into a filament. The areas are computed from the transported boundaries; grey is the separatrix.

With canonical brackets, Leibniz and the chain rule give $\dot{Q}_i = \{Q_i, H\} = \partial H / \partial P_i$ and $\dot{P}_i = -\partial H / \partial Q_i$: Hamilton's equations keep their form.

Generating functions

A transformation given as a formula is easy to check. The problem in practice runs the other way: to *produce* canonical transformations adapted to a given H . Generating functions do this, and the form language says in one line why they work.

Proposition 3.16 (Why generating functions work). *Let $S(q, P)$ be a function of n old positions and n new momenta, and define $(q, p) \mapsto (Q, P)$ implicitly by*

$$p_i = \frac{\partial S}{\partial q_i}, \quad Q_i = \frac{\partial S}{\partial P_i}, \quad (3.10)$$

wherever (q, P) are good coordinates. Then (3.9) holds.

Proof. In the coordinates (q, P) , (3.10) says exactly that

$$\sum_i p_i dq_i + \sum_i Q_i dP_i = dS.$$

Apply d . The right-hand side gives $ddS = 0$ (Theorem 2.13); the left gives $\sum dp_i \wedge dq_i + \sum dQ_i \wedge dP_i$. So $\sum dQ_i \wedge dP_i = -\sum dp_i \wedge dq_i = \sum dq_i \wedge dp_i$. ■

That is the mechanism in full: *a generating function is a primitive for a combination of canonical 1-forms, and $d \circ d = 0$ turns “exact” into “canonical”*. For example $S = \sum_i q_i P_i + f(q)$ gives $Q = q$ and $P = p - \nabla f$: shifting every momentum by a gradient is canonical. The function $\sum_i q_i P_i$ alone gives the identity.

Reading

The other classical types work identically. A *type-1* function $S(q, Q)$ with $p = \partial S / \partial q$ and $P = -\partial S / \partial Q$ says $p \cdot dq - P \cdot dQ = dS$; for instance $S = qQ$ gives $(Q, P) = (p, -q)$, which exchanges position and momentum up to a sign. Type 1 cannot generate the identity, whose q and Q are not independent, which is one reason type 2 is the workhorse and the type Lecture 5 uses.

Hamilton–Jacobi, recalled in this language

Hamilton–Jacobi looks for a type-2 generating function adapted to H : an $S(q, P)$ with

$$H\left(q, \frac{\partial S}{\partial q}\right) = K(P) \quad (3.11)$$

for some function K of the new momenta alone. The new equations are then $\dot{P}_i = -\partial K / \partial Q_i = 0$ and $\dot{Q}_i = \partial K / \partial P_i$, a constant: conserved momenta, and positions advancing at constant speed. Lecture 5 does this with $P = I$ the actions and $Q = \varphi$ the angles. With P held fixed, (3.11) also says that the n -dimensional set $\{(q, \partial S / \partial q)\}$ lies inside a level set of H — § 3.6 returns to this.

3.5 The Liouville volume $\omega^n / n!$

Proposition 3.17 (The Liouville volume form). *In canonical coordinates,*

$$\frac{\omega^n}{n!} = dq_1 \wedge dp_1 \wedge dq_2 \wedge dp_2 \wedge \cdots \wedge dq_n \wedge dp_n, \quad (3.12)$$

which as a measure is Lecture 1’s phase-space volume $dq_1 \cdots dq_n dp_1 \cdots dp_n$. In particular ω^n is nowhere zero, on any symplectic manifold, since a Darboux chart exists around every point.

Proof. Expanding ω^n by bilinearity, a term with a repeated index repeats a 1-form and vanishes. The survivors are the $n!$ orderings of the 2-forms $dq_i \wedge dp_i$, which commute by (2.9) and so are all equal. Reordering into $dq_1 \wedge \cdots \wedge dq_n \wedge dp_1 \wedge \cdots \wedge dp_n$ costs a sign $(-1)^{n(n-1)/2}$, which a measure does not see. ■

Theorem 3.18 (Liouville, in one line · as Proposition 3.15). *For any region R and any t for which φ_t is defined on all of R , $\text{vol}(\varphi_t(R)) = \text{vol}(R)$, where vol is the measure of $\omega^n / n!$.*

Proof. By [Proposition 3.15](#) and the rule $F^*(\alpha \wedge \beta) = F^*\alpha \wedge F^*\beta$ (2.15), $\varphi_t^*(\omega^n/n!) = (\varphi_t^*\omega)^n/n! = \omega^n/n!$. Now change variables in the integral, φ_t being an orientation-preserving diffeomorphism onto its image: $\int_{\varphi_t(R)} \omega^n/n! = \int_R \varphi_t^*(\omega^n/n!) = \int_R \omega^n/n!$. ■

This is [Theorem 1.17](#), now a consequence of the stronger $\varphi_t^*\omega = \omega$. For $n = 1$ the two coincide; for $n \geq 2$ the symplectic statement is strictly stronger.

Theorem 3.19 (Poincaré's integral invariant · proved modulo Stokes). *Let γ be a closed loop in $\mathbb{R}^{2n}$ and let φ_t be defined on it. Then*

$$\oint_{\varphi_t \circ \gamma} p \cdot dq = \oint_{\gamma} p \cdot dq.$$

The proof, Stokes on the tube the loop sweeps out ([Figure 3.4](#)), is in the reading.

Proof (★). Sweep the loop along the flow: $\Phi(s, u) = \varphi_s(\gamma(u))$ maps the cylinder $C = [0, t] \times S^1$ into phase space, with $\partial_s \Phi = X_H$. Orient C by $ds \wedge du$; its induced boundary is then the top loop minus the bottom one, so Stokes ([Theorem 2.23](#)) applied on C , as in [Corollary 2.26](#), gives

$$\oint_{\varphi_t \circ \gamma} p \cdot dq - \oint_{\gamma} p \cdot dq = \int_C \Phi^* d(p \cdot dq) = - \int_C \Phi^* \omega,$$

using $d(p \cdot dq) = \sum dp_i \wedge dq_i = -\omega$. Relative to $ds \wedge du$ the integrand $\Phi^* \omega$ has coefficient $\omega(\partial_s \Phi, \partial_u \Phi) = \omega(X_H, \partial_u \Phi) = dH(\partial_u \Phi) = \partial_u(H \circ \Phi)$, by (3.5). At each fixed s this is the derivative of a periodic function of u , so its integral over $u \in S^1$ vanishes; by Fubini, so does $\int_C \Phi^* \omega$. ■

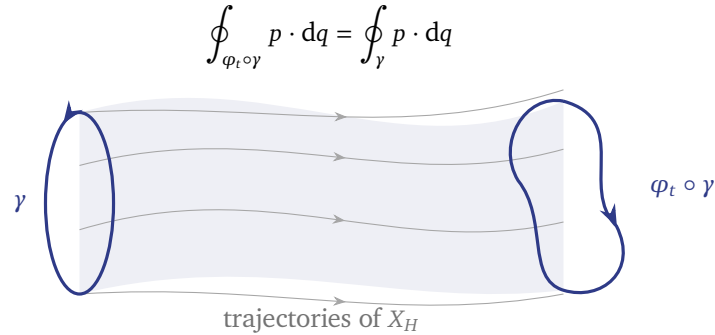


Figure 3.4: Poincaré's integral invariant: the loop is carried along the trajectories and deformed, and $\oint p \cdot dq$ does not change. By Stokes the change is $-\int \omega$ over the tube, and one edge of every cell of the tube is X_H , with $\omega(X_H, \cdot) = dH$, which integrates to zero around each loop.

For $n = 1$ this is Liouville again; for $n \geq 2$ it conserves the *sum of the signed areas of the loop's n shadows*, counted with winding number (a figure-eight shadow contributes the difference of its lobes), not a volume. It is the prototype of the action variable: [Lecture 5](#) integrates $p \cdot dq$ around a cycle on an invariant torus, and the flow carries the cycle to another with the same integral.

Theorem 3.20 (Gromov non-squeezing · cited). *A symplectomorphism can map a ball of radius r in $\mathbb{R}^{2n}$ into the cylinder $\{q_1^2 + p_1^2 < R^2\}$ only if $r \leq R$. (Gromov, Invent. Math. **82**, 1985.)*

For $n \geq 2$ the cylinder has infinite volume, so this is symplectic, not volumetric — and not local, since it concerns a ball of definite size. We shall not use it.

Reading

The cross-section must be a (q_1, p_1) -plane. Over the (q_1, q_2) -plane the symplectic map $(q, p) \mapsto (\varepsilon q, p/\varepsilon)$ squeezes a ball of radius r into $\{q_1^2 + q_2^2 < \varepsilon^2 r^2\}$, as thin as you like.

3.6 Lagrangian submanifolds

How large can a subspace be on which ω vanishes?

Let $\dim V = 2n$ with ω non-degenerate. The ω -complement of a subspace U is $U^\omega = \{v : \omega(v, u) = 0 \text{ for all } u \in U\}$; by (3.1) it is the preimage of the covectors vanishing on U , so

$$\dim U^\omega = 2n - \dim U. \quad (3.13)$$

Unlike an orthogonal complement, U^ω can contain U , since every vector is ω -orthogonal to itself. Call U *isotropic* if ω vanishes on it, i.e. $U \subseteq U^\omega$; then (3.13) gives $\dim U \leq 2n - \dim U$: *an isotropic subspace has dimension at most n .*

Definition 3.21 (Lagrangian subspace, Lagrangian submanifold). A *Lagrangian subspace* is an isotropic subspace of the maximal dimension n . A submanifold $L \subset \mathcal{M}$ is *Lagrangian* if it has dimension n and $\omega(v, w) = 0$ for all v, w tangent to L — equivalently, if the pullback of ω to L vanishes, written $\omega|_L = 0$.

This bound is why [Lecture 4](#) asks for exactly n integrals.

Proposition 3.22 (At most n commuting integrals). If $F_1, \dots, F_k$ satisfy $\{F_i, F_j\} = 0$ and $dF_1, \dots, dF_k$ are linearly independent at a point, then $k \leq n$.

Proof. At that point the vectors X_{F_i} are linearly independent, since $\iota_{X_{F_i}} \omega = dF_i$ and $v \mapsto \iota_v \omega$ is an isomorphism. They span an isotropic subspace, since $\omega(X_{F_i}, X_{F_j}) = \{F_i, F_j\} = 0$ by (3.6). So $k \leq n$. ■

Seeing it in $\mathbb{R}^4$

In $\mathbb{R}^4$, $\omega(v, w) = A_1 + A_2$, the signed areas of the two shadows of the parallelogram on v, w . A plane is Lagrangian when the two shadow areas of a basis parallelogram *cancel* ([Figure 3.5](#)).

Examples

Example 3.23 (Curves in the plane). For $n = 1$ every curve is Lagrangian: it has dimension $1 = n$, and a 2-form vanishes on parallel vectors. The notion has content from $n = 2$ on.

Example 3.24 (Positions and momenta). In $\mathbb{R}^{2n}$ the q -space $\{p = 0\}$ and each p -space $\{q = \text{const}\}$ are Lagrangian, since ω pairs a q -direction only with a p -direction. The (q_1, p_1) -plane is not: ω restricts to $dq_1 \wedge dp_1$ on it.

The most important examples are graphs, and they bring back [Lecture 2](#).

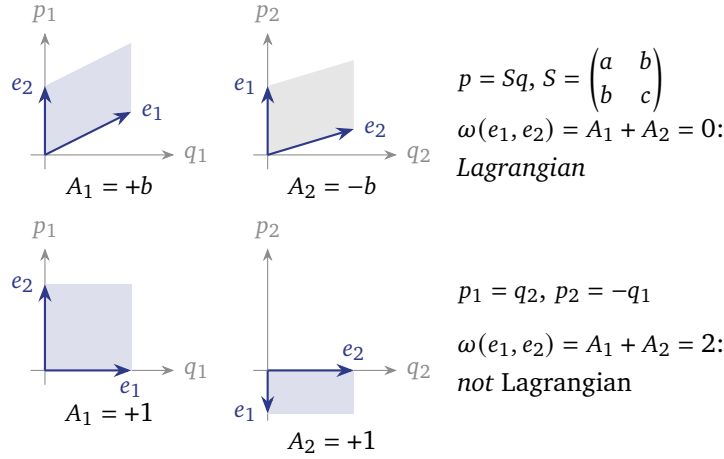


Figure 3.5: Two planes in $\mathbb{R}^4$, each drawn through the shadows of a basis in the two symplectic planes; coordinates are ordered (q_1, q_2, p_1, p_2) . Top: $e_1 = (1, 0, a, b)$, $e_2 = (0, 1, b, c)$ span the graph of a symmetric matrix (drawn with $a = 0.5$, $b = 0.8$, $c = 0.3$); the shadow areas are $+b$ and $-b$ and cancel. Bottom: $e_1 = (1, 0, 0, -1)$, $e_2 = (0, 1, 1, 0)$ span the graph of an antisymmetric one; both shadows run anticlockwise and the areas add. ω only ever looks at these two shadows.

Proposition 3.25 (Lagrangian graphs are closed 1-forms). Let $\sigma = \sum_i \sigma_i(q) dq_i$ be a 1-form on an open set of q -space, and $L_\sigma = \{(q, p) : p_i = \sigma_i(q)\}$ its graph. Then $\omega|_{L_\sigma} = -d\sigma$. Hence L_σ is Lagrangian if and only if σ is closed. In particular the graph of dS ,

$$p_i = \frac{\partial S}{\partial q_i}, \quad (3.14)$$

is Lagrangian for every function S .

Proof. Parametrise L_σ by q . Pulling back is substituting $p_i = \sigma_i(q)$:

$$\omega|_{L_\sigma} = \sum_i dq_i \wedge d\sigma_i = \sum_{i,j} \frac{\partial \sigma_i}{\partial q_j} dq_i \wedge dq_j = - \sum_{i,j} \frac{\partial \sigma_i}{\partial q_j} dq_j \wedge dq_i = -d\sigma,$$

by (2.6) and (2.10). For $\sigma = dS$ this vanishes by Theorem 2.13. ■

S is called a *generating function* of the Lagrangian submanifold, and the name is not a coincidence: by (3.11), a solution of Hamilton–Jacobi is a family of Lagrangian graphs, one for each value of P , each lying in a level set of H . Lecture 5’s $S(q, I)$ is exactly that, with the invariant tori as the family.

Reading

By the Poincaré lemma a closed σ is locally some dS ; globally it need not be. On the pendulum’s cylinder, with angle q , a rotation orbit at energy $E > 1$ is the graph $p = \sqrt{2(E + \cos q)}$ of the 1-form $\sigma = \sqrt{2(E + \cos q)} dq$ — closed, as every 1-form on a circle is, and Lagrangian. It is not dS for any function S on the circle, for the reason of Proposition 2.17: $\oint \sigma > 0$. Lecture 5 calls $\frac{1}{2\pi} \oint \sigma$ the rotating pendulum’s action. A nonzero action is precisely the obstruction to a global generating function, and Lecture 5 lives in that gap.

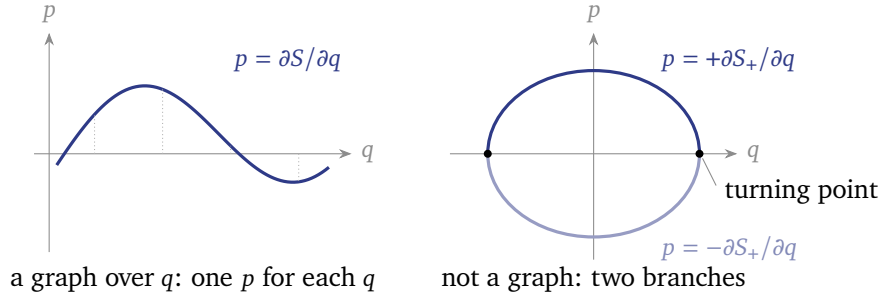


Figure 3.6: Left: the graph of dS , Lagrangian for every S (Proposition 3.25). Right: an oscillator orbit, Lagrangian by Example 3.23 but not a graph — two points over each q , and a vertical tangent at the turning points. Each branch is a graph, generated by $\pm S_+$; the turning points are where Lecture 5's $S(q, I)$ changes branch.

Example 3.26 (A Lagrangian torus in $\mathbb{R}^4$). Take the two-dimensional oscillator's integrals $F_i = \frac{1}{2}(q_i^2 + p_i^2)$ and, for $c_1, c_2 > 0$, the level set

$$T = \{F_1 = c_1, F_2 = c_2\} = \{q_1^2 + p_1^2 = 2c_1\} \times \{q_2^2 + p_2^2 = 2c_2\},$$

a circle in each symplectic plane: a torus (Figure 3.7). Its tangent spaces are spanned by $X_{F_1} = p_1\partial_{q_1} - q_1\partial_{p_1}$ and $X_{F_2} = p_2\partial_{q_2} - q_2\partial_{p_2}$, each tangent to its own circle and nonzero there, and $\omega(X_{F_1}, X_{F_2}) = \{F_1, F_2\} = 0$. So T is Lagrangian: each tangent parallelogram has one edge in each symplectic plane, so both its shadows are degenerate.

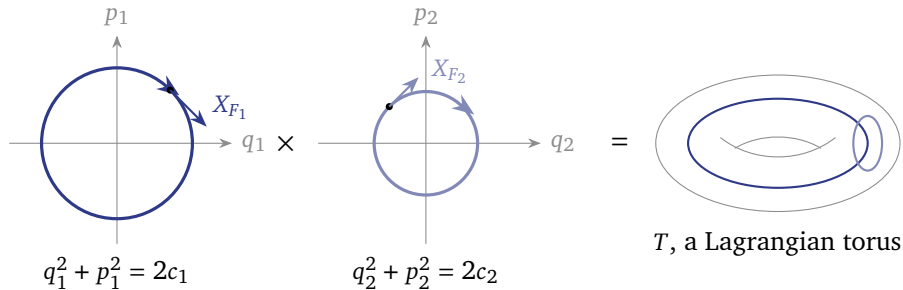


Figure 3.7: A level set of the 2D oscillator: a product of two circles, one in each symplectic plane, with X_{F_1}, X_{F_2} tangent and ω -orthogonal. So it is Lagrangian. Lecture 4 proves that this is no accident: each compact connected component of a regular level set of n independent commuting integrals is a Lagrangian torus.

Example 3.26 is Lecture 4's first proof step, done by hand; the general argument is Proposition 3.22's, with equality. Like the circle of Figure 3.6, T is not a graph over (q_1, q_2) .

3.7 Cotangent bundles and the canonical 1-form

Positions and their momenta

Let Q be a configuration space, a manifold of dimension n — the circle for the pendulum, the torus T^2 for the double pendulum. Its *cotangent bundle* is the set of pairs

$$T^*Q = \{(q, p) : q \in Q, p \in T_q^*Q\}, \quad \pi(q, p) = q,$$

a point of configuration space and a covector there. A chart $q_1, \dots, q_n$ on Q writes the covector as $p = \sum_i p_i dq_i$ (Definition 2.1), and (q, p) is then a chart on the $2n$ -dimensional T^*Q . That momenta are covectors shows in a change of chart $q' = \phi(q)$: writing the same covector in both bases, $\sum_j p'_j dq'_j = \sum_i p_i dq_i$, forces

$$p_i = \sum_j p'_j \frac{\partial q'_j}{\partial q_i} : \quad (3.15)$$

momenta transform like the components of a gradient, not like those of a velocity. For polar coordinates on the plane, (3.15) gives $p_r = (x p_x + y p_y)/r$ and $p_\theta = x p_y - y p_x$ — the momentum conjugate to the angle is the angular momentum, as it should be.

The canonical 1-form

Definition 3.27 (Canonical 1-form). At a point (q, p) of T^*Q and for a tangent vector v there,

$$(p \cdot dq)_{(q,p)}(v) = p(d\pi v). \quad (3.16)$$

In words: push v down to configuration space, and feed it to the covector p — which is part of the point, and so is already there (Figure 3.8).

In a chart, $d\pi v = \sum_i v^{q_i} \partial_{q_i}$ drops the momentum components of v , so (3.16) is $(\sum_i p_i dq_i)(v)$. The form of Example 2.4, and of Lecture 5's action integral, is therefore *the same 1-form in every cotangent chart*, because (3.16) mentions no chart.

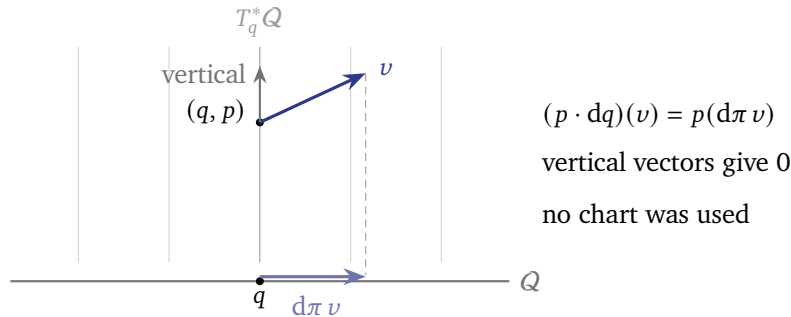


Figure 3.8: The canonical 1-form. Above each q sits the fibre T_q^*Q of momenta there. A vector v at (q, p) is pushed down to $d\pi v$ and fed to p , which is part of the point. A vector along the fibre projects to zero and is not seen.

Theorem 3.28 (Every cotangent bundle is symplectic). $\omega = -d(p \cdot dq)$ is a symplectic form on T^*Q , and in every cotangent chart it is $\sum_i dq_i \wedge dp_i$.

Proof. In a chart, $d(\sum_i p_i dq_i) = \sum_i dp_i \wedge dq_i = -\sum_i dq_i \wedge dp_i$, so ω is the standard form there, which is non-degenerate (Example 3.3). It is closed because it is exact: $d\omega = -dd(p \cdot dq) = 0$. And it is the same form in every chart, because $p \cdot dq$ is. ■

So the pendulum's cylinder, the double pendulum's T^*T^2 and the rigid body's $T^*SO(3)$ are symplectic, with no choice made; and Poincaré's integral invariant holds on any cotangent bundle, since its proof used only $d(p \cdot dq) = -\omega$ and $\iota_{X_H}\omega = dH$.

The minus sign, and why it costs nothing

The minus sign in $\omega = -d(p \cdot dq)$ is forced by the wedge order $dq \wedge dp$, which (3.4) showed is the one giving Hamilton's equations. Sources that write $\omega = dp \wedge dq$ print $\omega = d(p dq)$; do not transcribe them.

The minus is compensated where it matters. For the oscillator $H = \frac{1}{2}(p^2 + \nu^2 q^2)$, $\dot{p} = -\nu^2 q$, so at $(q, p) = (1, 0)$ the motion is straight down: orbits in the (q, p) -plane run *clockwise*. A clockwise orbit γ is the boundary of the enclosed disc D with the *opposite* of the orientation that $dq \wedge dp$ gives it, so by Stokes

$$\oint_{\gamma} p \, dq = - \int_D d(p \, dq) = + \int_D \omega = \text{area of } D = \frac{2\pi E}{\nu} > 0.$$

Two minus signs, one from the wedge order and one from the direction of motion, and the action integral of [Lecture 5](#), taken along the motion, comes out positive.

Reading

Point transformations are canonical

A diffeomorphism ϕ of configuration space moves momenta by the transpose of its inverse Jacobian — that is (3.15) read as a map — and so lifts to a map $\hat{\phi}(q, p) = (\phi(q), p')$ of T^*Q , the *cotangent lift*. Because $p \cdot dq$ is defined without a chart, $\hat{\phi}^*(p \cdot dq) = p \cdot dq$, and hence $\hat{\phi}^*\omega = \omega$ by (2.15). Every change of configuration coordinates, with the momenta carried along as covectors, is a canonical transformation. This is why passing to polar coordinates, with p_θ the angular momentum, never needs checking. It is also the type-2 generating function $S(q, p') = \sum_j p'_j \phi_j(q)$ of [Proposition 3.16](#), as a one-line computation with (3.10) confirms.

Summary.

- *The structure.* A closed, non-degenerate 2-form ([Definition 3.2](#)); non-degeneracy turns covectors into vectors (3.1) and forces even dimension ([Proposition 3.4](#)). The standard form is $\sum dq_i \wedge dp_i$, in that order, because that order gives Hamilton's equations (3.4).
- *One structure.* $\iota_{X_f}\omega = df$ recovers [Lecture 1](#)'s X_f , with $\omega(X_f, X_g) = \{f, g\}$ ([Proposition 3.6](#)), and the bracket satisfies Jacobi exactly when $d\omega = 0$ ([Theorem 3.8](#)).
- *Darboux.* Locally every symplectic manifold is standard ([Theorem 3.10](#)), even with n given commuting functions as momenta ([Theorem 3.12](#)). Everything interesting is global.
- *Canonical transformations* are the symplectomorphisms; Hamiltonian flows are among them ([Proposition 3.15](#)); generating functions work because $d \circ d = 0$ ([Proposition 3.16](#)). The flow preserves the volume $\omega^n/n!$ and Poincaré's $\oint p \cdot dq$ ([Theorems 3.18](#) and [3.19](#)).
- *Lagrangian submanifolds:* dimension n , $\omega|_L = 0$. At most n independent functions commute ([Proposition 3.22](#)); graphs of dS and the oscillator torus are Lagrangian ([Proposition 3.25](#), [Example 3.26](#)).
- *Cotangent bundles* are symplectic for free, with $\omega = -d(p \cdot dq)$ ([Theorem 3.28](#)): the minus is forced by the wedge order and cancelled by the clockwise motion of orbits.

Tutorial. Exercise Sheet 3.

References

da Silva, *Lectures on Symplectic Geometry*, §§1–4 (symplectic forms, cotangent bundles and the tautological form, Lagrangian submanifolds, generating functions) and §§7–8 (the Moser trick and Darboux); Arnold, *Mathematical Methods of Classical Mechanics*, §§37–38 and 40–41 (symplectic structures, integral invariants, the Poisson bracket from ω), §43 (Darboux), §44 (the integral invariant of Poincaré–Cartan) and §§47–48 (Hamilton–Jacobi, generating functions); Bolsinov and Fomenko, *Integrable Hamiltonian Systems*, §1.3 (Darboux, and its version adapted to commuting functions). **Conventions.** da Silva’s agree with ours exactly — $\omega = -d\alpha$ with α the tautological form, $\iota_{X_H}\omega = dH$, $\{f, g\} = \omega(X_f, X_g)$, and the engine’s minus sign. Arnold and Bolsinov–Fomenko write $\omega = \sum dp \wedge dq$, so their ω is minus ours: check any formula from them against (3.4) and Theorem 3.28 before transcribing it.